

Anti-Involution and Global Price Pressure: China's New Supply-Side Reform and the Spillovers of Green Industrial Overcapacity

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1 Introduction and the Policy Engine

China accounts for more than 70% of global electric-vehicle production (IEA, 2025a) and holds more than 80% of manufacturing capacity across key solar PV supply-chain stages. A single policy apparatus produced both positions. The state channels capital and labor toward centrally designated strategic industries, and in doing so it has created globally competitive firms. It has also generated chronic overcapacity and persistent domestic price wars, both of which now spill into trade friction.

I argue that the anti-involution campaign central authorities announced in 2025 is a supply-side reform *within* the industrial-policy state, not a retreat from it. Anti-involution targets the destructive price competition that follows from subsidized overentry, but it does not address the demand-side imbalance that makes overentry profitable in the first place. Without complementary demand-side reform, anti-involution may discipline the worst domestic price wars while the global economy continues to absorb Chinese overcapacity as downward pressure on traded-goods prices.

The sequence begins with central-government target-setting. Programs such as Made in China 2025 and the designation of the “new three” (electric vehicles, lithium-ion batteries, and solar PV) as national champions signal where state resources will flow. Local governments then compete to attract investment in these sectors. Fang et al. (2025b) find that this competition operates through a tournament mechanism: provincial and municipal officials, whose career advancement depends on measurable economic performance, offer subsidized land and cheap credit to firms willing to locate within their jurisdictions. Tax holidays and purpose-built industrial parks round out the package. The central government sets objectives; local governments compete on implementation, producing parallel investment across dozens of provinces that each build capacity in the same industries.

Soft budget constraints prevent market discipline from correcting the resulting oversupply. In a hard-budget-constraint environment, losses force exit and bankruptcy clears the surplus. In China, state-controlled banks and local government financing vehicles absorb losses instead, because lenders face political incentives to extend credit rather than recognize write-downs. The clearing mechanism is blunted. Firms that would otherwise fail keep operating, and new entrants keep arriving, because the cost of entry is socialized while the political reward for attracting investment remains private to the local official who secured it. The result is predictable: subsidized expansion without proportional gains in efficiency. Branstetter and Li (2022) find exactly this pattern for Made in China 2025 targets, and Branstetter et al.

(2023) confirm it for Chinese industrial subsidies more broadly. Subsidized firms invest more. They do not consistently become more productive.

The final link is the gap between output and domestic absorption. China's household consumption share of GDP remains low by international standards, not because Chinese households choose austerity but because the growth model systematically channels income toward investment. High savings rates, sustained in part by weak social insurance and the post-2021 property adjustment, compress domestic purchasing power further. The implication for industrial policy is direct: every unit of capacity added to a targeted sector must find a buyer either at home or abroad. When domestic demand is structurally insufficient, the surplus generates price wars at home and export pressure abroad. The two outcomes are not separate problems; they are the same problem viewed from different sides of the border.

Policy diffusion reinforces the pattern. Fang et al. (2025a) document that local governments systematically imitate peer jurisdictions' industrial policies, duplicating capacity across hundreds of cities. The resulting subsidies have trade consequences: Rotunno and Ruta (2024) find that Chinese industrial subsidies are associated with increased export volumes in subsidized sectors.

If this account is right, overcapacity should concentrate in centrally targeted sectors. The “new three” provide the test case.

2 Overcapacity in the “New Three”

Table 1 summarizes capacity pressure across China's three flagship clean-technology sectors. In each, installed capacity exceeds plausible demand and output prices have fallen sharply.

Solar PV. Solar module prices collapsed between 2023 and 2025. The arithmetic behind the collapse is straightforward: China's solar photovoltaic manufacturing capacity exceeds 1 TW per year, while global installations reached roughly 600 GW in 2024 (IEA, 2025b). Even if every panel installed anywhere in the world came from a Chinese factory, nearly half of Chinese capacity would sit idle. Chinese firms account for more than 80% of global output at every major stage of the supply chain, from polysilicon and wafer production through cell and module assembly. This dominance was not built on cost advantages alone. The OECD's MAGIC database identifies solar as the most heavily subsidized sector it tracks: government support to solar firms averaged 3.2% of revenue, compared with 0.9% across all sectors in the database (OECD, 2026, 2025). The subsidy wedge explains why capacity continued to expand even as prices fell below the point where unsubsidized entry would have stopped.

Lithium-ion batteries. China hosts about 85% of global battery cell manufacturing capacity (USCC, 2025; IEA, 2025a). The investment thesis behind this expansion is that global battery demand will more than triple by 2030, and firms building capacity now will capture that future market. The thesis may prove correct in the long run, but it does not solve the near-term problem. Battery pack prices in China fell about 30% in 2024, compared with 10–15% in Europe and the United States (IEA, 2025a). If the price decline reflected a global technology-driven cost curve, the drop would be roughly uniform across markets. The fact that Chinese prices fell roughly twice as fast as Western ones points to a domestic glut, not shared progress down the learning curve.

Electric vehicles. The EV sector presents the most politically visible case. When national purchase subsidies ended in early 2023, Chinese EV makers launched successive rounds of price cuts, each intended

to gain market share at rivals’ expense. The price war has not stopped. NBS data show that automobile manufacturing PPI fell 2.8% in 2025, slightly worse than the 2.6% decline for industry as a whole (NBS, 2026). Global EV sales exceeded 17 million units in 2024, but Chinese production capacity is widely reported to exceed even the combined total of domestic absorption and exports (IEA, 2025a; Mayr et al., 2026). The European Parliament study documents a rising share of loss-making firms in the automobile sector (Mayr et al., 2026). In a competitive market, persistent losses lead to exit. In China’s EV sector, the soft budget constraint described in Section 1 keeps loss-making firms alive, and the price floor that exit would establish never materializes.

Sector	China-side indicator	Demand benchmark	Price/margin symptom
Solar PV	Mfg. capacity >1 TW/yr	~600 GW installed (2024)	Sharp price decline
Li-ion batteries	~85% of global mfg. capacity	Below installed capacity	Pack prices fell ~30% in China (2024)
EVs	Capacity exceeds demand	Global sales >17M (2024)	Auto mfg. PPI –2.8% (2025)

Table 1: Descriptive indicators of capacity pressure in China’s “new three” sectors. Solar: IEA (2025b); OECD (2026). Batteries: USCC (2025); IEA (2025a). EVs: IEA (2025a); NBS (2026).

The definition of overcapacity is contested. Beijing maintains that current capacity is forward-looking: global green demand will eventually absorb what today appears surplus. The OECD and IMF treat the gap as structural overcapacity rooted in the subsidy architecture described in Section 1, arguing that policy-driven entry has decoupled investment from market signals (IMF, 2026; OECD, 2025). I follow the structural interpretation for two reasons. First, the soft budget constraint means that capacity investment reflects the incentives of local officials and state lenders, not any firm’s assessment of future demand. The “forward-looking” label implies a market rationality that the tournament mechanism systematically overrides. Second, even on a forward-looking reading, financing the interim capacity gap imposes real costs on credit allocation and on competing producers forced to exit.

Across all three sectors the story is the same: capacity has outrun demand, and margins have compressed to the point where firm survival depends on continued state support rather than commercial viability. The question is what Beijing intends to do about it.

3 Anti-Involution as Domestic Response

3.1 The policy and its instruments

“Involution-style competition” (*neijuan shi jingzheng*) describes a pattern in which firms compete on price and scale alone, compressing margins without generating productivity gains. No firm gains a durable advantage because competitors match each price cut: the equilibrium is a race to the bottom in which every participant loses margin and no one gains market power. The result is value destruction rather than creative destruction. Anti-involution is Beijing’s label for this pathology, and its policy response.

Central authorities moved to address involution through administrative intervention in 2025. In July, a State Council executive meeting on the new-energy vehicle industry called for curbing “irrational competition” and announced strengthened cost surveys and price monitoring across the NEV supply chain (State Council, 2025a). In January 2025, the NDRC issued guidelines for building a unified national market, targeting local protectionism and regional market barriers that fragment domestic demand and enable duplicated capacity (State Council, 2025b). Specific instruments include tighter safety and

quality standards (revised EV standards take effect in 2026) and merger facilitation. Local-government performance evaluation is to shift from output quantity toward quality metrics. Each instrument operates within the existing industrial-policy apparatus: the state is regulating competition, not marketizing it.

3.2 The MITI analogy and its limits

A serious counterargument holds that anti-involution represents genuine policy learning: a maturation from quantity-maximizing industrial policy toward quality-focused upgrading. The analogy to Japan's MITI is tempting. During the 1960s and 1970s, MITI used administrative guidance to curb "excessive competition" (*kato kyoso*) in steel and automobiles, facilitating consolidation that contributed to subsequent technological upgrading (Johnson, 1982).

But did MITI succeed because of administrative coordination alone? Japan's industrial consolidation coincided with one of the fastest sustained increases in household living standards in economic history. Wage-bargaining institutions spread the gains of industrialization, and rising domestic incomes absorbed much of what industry produced. The demand side of the economy was doing the heavy lifting: MITI could consolidate supply because demand was growing fast enough to make the consolidated firms profitable. China faces the opposite. Household consumption remains around 39–40% of GDP, below the average for upper-middle-income economies and far below advanced-economy benchmarks (World Bank, 2025). The household savings rate sits at approximately 34%, against a global average of roughly 24% (Gunter et al., 2025). Financial repression and the systematic channeling of income toward investment compress the household share of national income. The property-sector adjustment since 2021 has eroded household wealth, and limited social protection reinforces precautionary saving. As Klein and Pettis (2020) argue, these are features of the growth model, not incidental frictions: the model suppresses consumption by design.

This is the core difficulty. Supply-side consolidation only works when the demand side is healthy enough to sustain the surviving firms at profitable scale. Anti-involution may slow the price war among EV manufacturers, but it cannot create the domestic buyers those manufacturers need. Consolidating producers accomplishes little if the consolidated firms still face a market too small to absorb their output. The textbook prescription would be to raise the household share of income through fiscal transfers and expanded social insurance, coupled with a relaxation of financial repression. None of these appear in the anti-involution policy toolkit.

Scale makes matters worse. MITI coordinated a handful of firms in a few sectors. China's overcapacity spans hundreds of cities and thousands of firms across multiple layers of government with conflicting objectives (Gunter et al., 2025). The coordination problem is not incrementally larger; it is qualitatively different. The fiscal architecture of the state also works against compliance. Local governments depend on land-transfer fees and industrial-project revenue for a large share of their discretionary budgets. The tournament mechanism documented in Section 1 persists because the fiscal incentive to attract investment persists. Anti-involution punishes the symptom (destructive price competition) without reforming the incentive structure that generates overcapacity. This creates a familiar principal-agent problem: Beijing issues the directive, but the agents who must implement it are the same local officials whose budgets depend on the investment the directive is trying to restrain. As long as local officials gain revenue and career advancement from attracting new capacity, compliance with anti-involution directives will be selective at best.

I therefore read anti-involution as an administrative correction, not a structural reform. It may reduce

the worst price wars through tighter standards and managed consolidation, but it does not address the root cause: production capacity that exceeds domestic demand, sustained by a growth model that favors investment over consumption. The surplus that anti-involution cannot absorb domestically does not disappear. It shows up in export markets.

4 Global Spillover: Downward Price Pressure

4.1 China Shock 2.0

When Autor et al. (2013) documented the original China Shock, the story was straightforward: WTO accession accelerated a trade shock rooted in comparative advantage in labor-intensive manufacturing, and American workers in competing industries paid the price. Two decades later, the shock has returned in a different form. What China now exports in volume is capital-intensive green technology: the “new three” of EVs, batteries, and solar PV. The goods have changed. The disruption has not.

I stress the continuity because the underlying mechanism is different enough to obscure it. China possesses genuine endowments in engineering talent and scale economies, but the policy apparatus described earlier amplified these advantages through directed capital allocation and protected domestic markets. The current shock is not purely market-driven. It combines real cost advantages with policy-amplified capacity expansion. The welfare calculus is more complex than in the original episode because the goods carry positive externalities: cheap solar panels and batteries slow climate change. The original China Shock asked whether cheap manufactures justified displaced workers. This one asks the same question with an additional complication.

Strategic entry versus residual exports

Not all Chinese green-tech exports reflect the same dynamic, and the distinction matters for policy. Some firms built export-oriented capacity strategically, planned before domestic margins collapsed. BYD’s plants in Thailand and Hungary are examples: deliberate market entry into new geographies, the kind of outward FDI that any capital-exporting economy would produce. Others are redirecting output abroad because the domestic price war left them unable to sell at home. This second category is closer to loss-tolerant export competition, where firms accept margins below long-run average cost and the soft budget constraint extends their survival horizon well beyond what private capital markets would finance.

Trade statistics do not distinguish between strategic entry and residual dumping. In practice, firms optimize across domestic and export markets simultaneously, and the same firm may be strategic in one market and residual in another. But the structural point holds regardless: when aggregate capacity exceeds domestic absorption, the export market functions as a pressure valve, and the pressure is downward on prices.

4.2 Evidence of downward price pressure

Chinese export volumes in clean-technology sectors have risen sharply since 2022 while unit values have fallen. The USCC documents this pattern across several sectors (USCC, 2025). Rotunno and Ruta (2024) provide econometric evidence that the relationship is systematic: Chinese industrial subsidies are associated with increased export volumes and depressed export prices in subsidized sectors. The direction of the effect is what the earlier sections of this essay would predict. Subsidies expand capacity beyond

what domestic demand can absorb, and the surplus enters world markets at prices that unsubsidized competitors find difficult to match.

A note on terminology. I avoid “exported deflation” because the phrase implies a general price-level effect. What is occurring is a decline in the relative price of clean-technology goods in world markets: a terms-of-trade improvement for importing countries but a deterioration for China and for competing producers elsewhere. The policy problem is sectoral adjustment, not macroeconomic deflation management.

4.3 The green-public-goods counterargument

The strongest objection to treating overcapacity as a problem runs as follows. Cheap Chinese solar panels and batteries accelerate global decarbonization, particularly in developing countries that cannot afford unsubsidized clean technology. On standard trade-theory grounds, a subsidy on goods with large positive externalities may be globally welfare-improving. For many countries in Sub-Saharan Africa and South Asia, affordable solar modules are the difference between decarbonized electrification and continued fossil-fuel reliance.

Yet the welfare case is less clean than this framing suggests. Adjustment costs fall unevenly: workers in European solar-module assembly or American EV-component manufacturing bear concentrated losses while benefits diffuse across consumers, replicating the distributional pattern that made the original China Shock politically corrosive. Dependence on a single subsidized source also creates supply-chain vulnerability, most acute at upstream chokepoints such as polysilicon refining where Chinese dominance is most pronounced. The trade-policy responses already deployed, meanwhile, are fragmenting the very markets through which welfare gains would flow. The United States imposed a 100% tariff on Chinese EVs in 2024 (USTR, 2024); the European Union adopted definitive countervailing duties ranging from 17.0% for BYD to 35.3% for SAIC (European Commission, 2024). In a first-best world, importing countries would accept subsidized goods and compensate displaced workers through domestic transfers. No major economy has chosen this path. The tariff responses actually observed are second-best policy driven by political-economy constraints: governments respond to concentrated producer losses rather than diffuse consumer gains. The irony is that the welfare benefit of subsidized overcapacity depends on open markets, and the overcapacity itself is provoking their closure. The subsidy that was supposed to accelerate decarbonization is instead accelerating trade fragmentation.

This leaves the system in an unstable position. Domestic demand cannot absorb the capacity, and export markets are fragmenting under tariff pressure. The question is whether China’s policy apparatus can reform the growth model that created the problem, or whether it will settle for managing the symptoms.

5 Tensions and Outlook

Anti-involution targets the wrong end of the problem. It can discipline the price competition that drives margins below zero, but insufficient domestic demand, the deeper issue, stays untouched. Raising quality thresholds and facilitating consolidation does not create buyers.

The most likely trajectory is a muddled middle. Solar manufacturing may see partial rationalization as small firms exit, but the EV sector is harder to consolidate because local governments hold stakes in regional automakers. No official wants to close a factory that employs constituents and generates tax revenue, even if the factory cannot turn a profit. Capacity additions already in the pipeline will arrive

even as demand growth slows. On the external side, trade responses will fragment the export market without eliminating it: the United States has opted for high tariff walls while much of the Global South remains open to Chinese imports on price grounds alone. Chinese firms will redirect exports toward whichever markets remain accessible.

What would genuine structural reform look like? It would require raising the household share of national income. That means expanding social insurance to reduce precautionary saving, and reforming the fiscal system so that local governments depend less on land sales and industrial project revenue. It also means allowing real interest rates to rise, so that household savings earn a market return rather than being channeled at repressed rates into industrial investment. Each reform is politically difficult because each redistributes income away from the state sector and toward households. Anti-involution asks none of these questions.

Yet the campaign does reveal something about the state's capacity for self-correction. The same apparatus that created globally competitive green-manufacturing sectors now attempts to regulate the overcapacity it produced. That is not retreat. It is the industrial-policy state turning its own tools inward, and the fact that Beijing recognizes the problem at all is itself informative. The question is whether recognition leads to reform or merely to management. The precedents are not encouraging: China has discussed rebalancing toward consumption since at least the Twelfth Five-Year Plan (2011–2015), and the household consumption share of GDP has barely moved.

I see two indicators that would undermine this assessment: a sustained rise in China's household consumption share of GDP, or convergence of capacity utilization in the “new three” toward international norms. Neither is visible in recent data.

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